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Car and Pedestrian Detection using YOLO – Performance Analysis Report

Introduction:

The objective of this project is to develop an object detection model capable of identifying pedestrians and cars in images using the YOLO (You Only Look Once) deep learning framework. In this project, the YOLOv8 (Y object detection model was trained on a custom dataset containing images of cars and pedestrians to perform multi-class object detection. and evaluated using standard performance metrics such as Precision, Recall and mAP.

Objective:

The objective of this assignment is to:

- Create a dataset containing pedestrians and cars
- Annotate the dataset in YOLO format
- Train a YOLOv8 object detection model
- Evaluate the performance of the trained model
- Analyze failure cases and suggest improvements:

Dataset Description:

A custom dataset containing images of pedestrians and cars was created by combining publicly available datasets and filtered images.

A custom dataset containing annotated images of cars and pedestrians was used for training and evaluation of the model.

- Total number of Source images: 931
- Training images(70%): 656
- Validation images(20%): 183
- Testing images(10%): 92
- Number of classes: 2 (Car and Pedestrian)

The dataset was divided using a standard 70:20:10 split ratio to ensure balanced training and evaluation of the mode

Some images in the dataset contain both cars and pedestrians within the same scene. This enables the model to learn multi-class object detection and improves its ability to detect both object classes simultaneously in real-world environments.

Annotation Format:

The dataset images were annotated using the YOLO-compatible object detection format. Each object instance (car or pedestrian) was labeled with a bounding box using normalized coordinates.

The annotation format used for each object is:

(class_id , x_center , y _center , width , height)

Where:

- class_id represents the object class (0 = car, 1 = pedestrian)
- x_center, y_center represent the normalized center coordinates of the bounding box
- width and height represent the normalized bounding box dimensions

All coordinate values were normalized with respect to image width and height, ensuring compatibility with the YOLOv8 detection model.

Data Preprocessing and Augmentation:

Before training the model, preprocessing steps were applied to standardize the dataset:

- Auto-orientation was applied to correct image rotation.
- All input images were resized to 640×640 pixels to maintain consistency during training.

Data augmentation was applied only to the training dataset to improve generalization, and prevent overfitting.

- Brightness adjustment (−10% to +10%)
- Gaussian blur (up to 1.2px)
- 3× augmentation

These augmentation techniques simulate real-world variations such as lighting changes and camera blur, helping the model learn robust object features.

After augmentation:

Dataset	**Images**
Training Images	1964
Validation Images	183
Test Images	92
Total Images	2239

Model Architecture and Training:

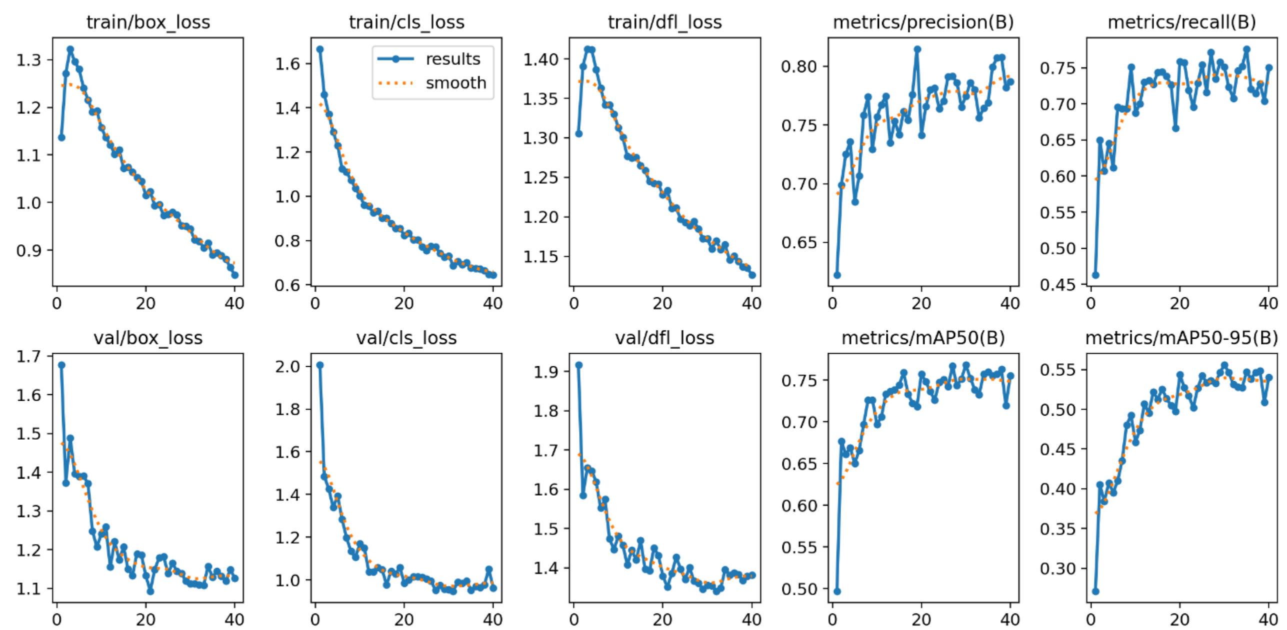
The **YOLOv8n (nano)** model was selected for this task due to its lightweight architecture and efficient performance for real-time object detection tasks.

The model was trained using the following parameters:

- Model: YOLOv8n
- Image size: 640 × 640
- Maximum epochs: 100
- Batch size: 8
- Early stopping: Enabled (patience = 10)

Early stopping was applied to prevent overfitting.

Training and Validation Performance:



- The training loss keeps decreasing, which shows that the model is learning properly.
- The validation loss also decreases and then becomes stable, ie, the model works well on unseen data.
- Precision increases during training, so the model makes fewer wrong detections.
- Recall also improves, which means the model detects more actual objects.
- The mAP@0.5 value increases and reaches around 0.76, showing good overall detection accuracy.
- The mAP@0.5:0.95 metric also improves, which indicates better object localization.
- After around 25–30 epochs, the graphs become stable, meaning the model has learned most of the important features.
- There are no large fluctuations in validation results, so overfitting is minimal.

Model Performance Metrics:

After training, the model was evaluated on the validation dataset using standard object detection metrics.

Overall Performance

- **Precision:** 0.777
- **Recall:** 0.751
- **mAP@50:** 0.769
- **mAP@50–95:** 0.556

The achieved **mAP@0.5 of 76.9%** indicates strong detection accuracy for both car and pedestrian classes. The balanced **Precision** and **Recall** values demonstrate effective object localization with minimal false positives and missed detections, highlighting the model's reliable performance on the validation dataset.

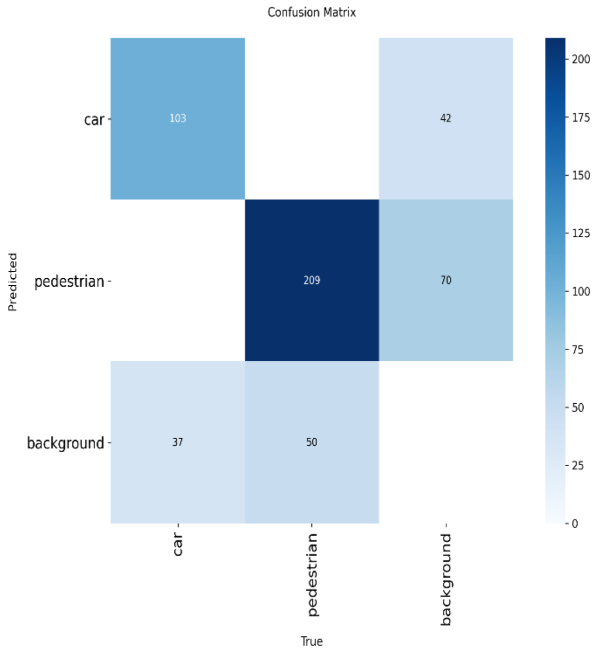
```
/precision(B)': 0.7772325575444919, 'metrics/recall(B)': 0.7513608027512906, 'metrics/mAP50(B)': 0.7693289013221373, 'metrics/mAP50-95(B)': 0.5559097900991532,
```

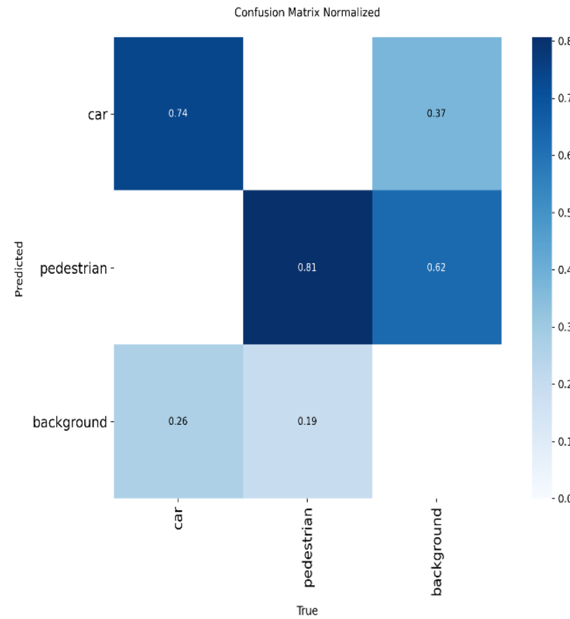
Class-wise Performance:

	Class	Precision	Recall	mAP@0.5:0.95
0	Car	0.757	0.734	0.561
1	Pedestrian	0.797	0.768	0.551

The class-wise performance results show that the model detects pedestrians slightly better than cars. This may be because pedestrians in the dataset were more clearly visible, while cars sometimes appeared smaller, partially hidden, or in complex backgrounds.

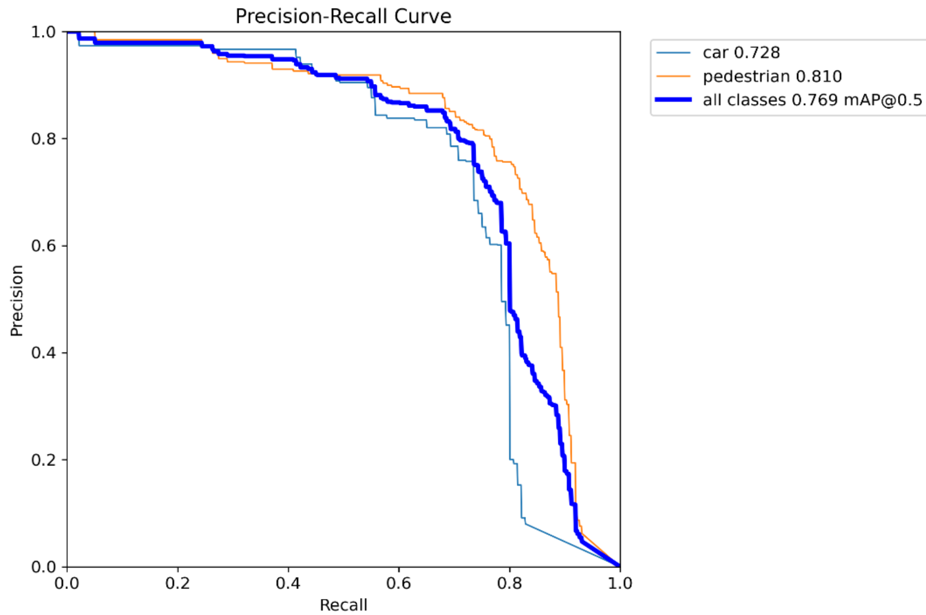
Confusion Matrix Analysis:





- The model correctly detected most cars and pedestrians, which shows that it has learned the object features well.
- Pedestrian detection performed slightly better than car detection, meaning the model is more confident and accurate in identifying human instances.
- Car detection was also strong, but a little lower compared to pedestrians, possibly because cars vary more in size and appearance.
- Some objects were missed and predicted as background. These missed detections mainly occurred when the objects were small, far away, or partially hidden.
- A small number of incorrect predictions occurred in complex scenes where the model confused objects or struggled with visual similarity.
- Overall, the model demonstrates stable and reliable performance, with only minor errors in challenging conditions.

Precision–Recall Curve:

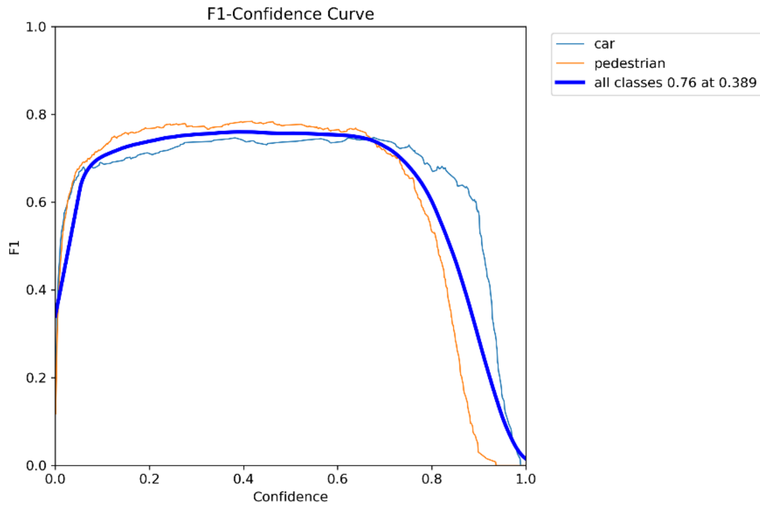


- **Precision (0.777):**

  Precision indicates how many of the detected objects were actually correct. A precision score of 0.777 means that most of the model's predictions were accurate, with only a few false detections.
- **Recall (0.751):**

  Recall measures how many of the actual objects present in the image were successfully detected. A recall score of 0.751 shows that the model was able to detect the majority of cars and pedestrians, although some objects were missed.
- The relatively balanced precision and recall values indicate that the model maintains a good trade-off between avoiding false positives and minimizing missed detections.
- Overall, these metrics suggest that the model demonstrates reliable detection performance with acceptable error rates.

F1 Score Analysis



- When the confidence level starts increasing, the F1 score also improves. This means the model becomes more balanced in detecting objects correctly.
- The highest F1 score is **0.76 at confidence 0.389**, which shows the best overall performance point.
- At this confidence value, the model maintains a good balance between detecting objects and avoiding wrong detections.
- If the confidence is set too high, the model becomes very strict and may miss some objects.
- If the confidence is too low, the model detects more objects but also increases false detections.
- Therefore, selecting the right confidence threshold is important for achieving better real-world performance.
- Overall, the model shows stable and reliable behavior around the optimal confidence value.

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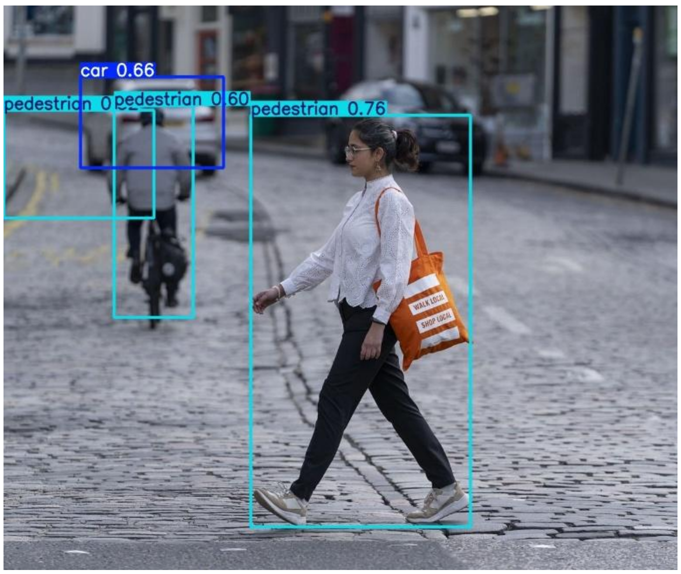
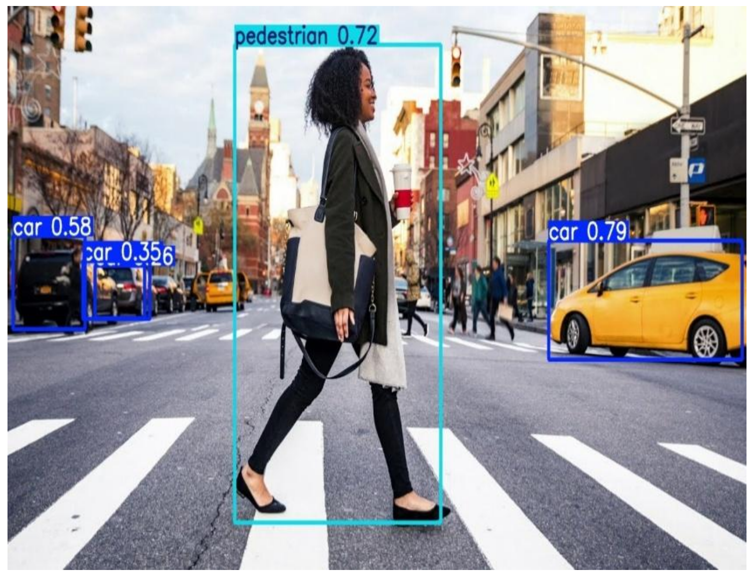
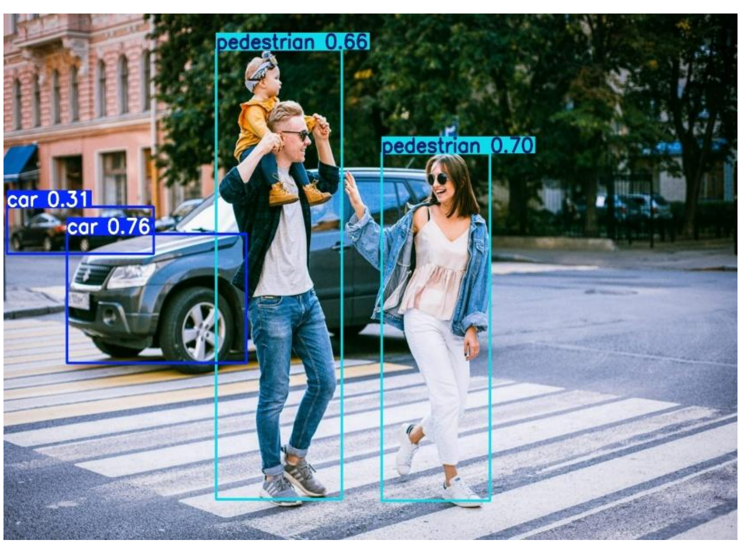
Detection Results on Test Data:

The trained model was evaluated on test images that were not used during training. The detection results show that the model is able to correctly identify most cars and pedestrians with properly placed bounding boxes. In clear and well-lit scenes, the model performs accurately and detects multiple objects effectively. However, in some challenging cases, such as small or partially hidden objects, a few detections were missed. Overall, the model demonstrates good performance and generalization on unseen test data

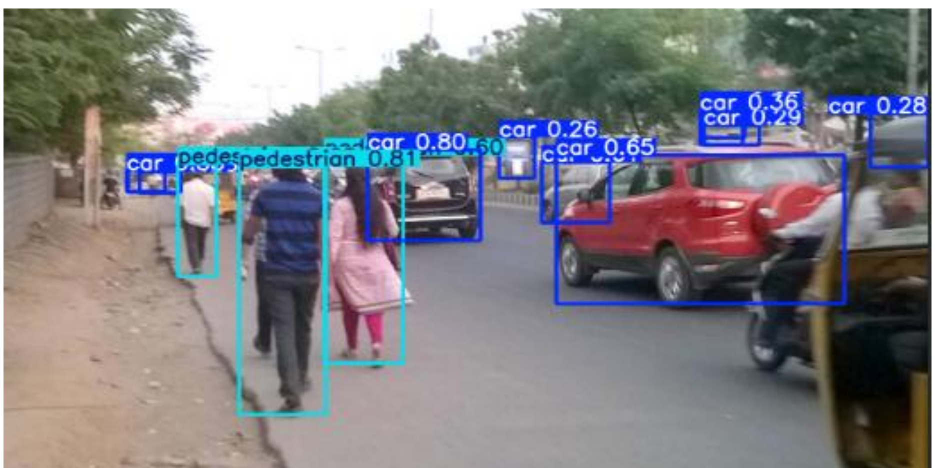






Real-World Image Evaluation:

The trained model was tested on unseen test images to evaluate its performance in real-world scenarios. The detection outputs demonstrate the model's ability to correctly identify cars and pedestrians in various environmental conditions.



Failure cases / issues observed:

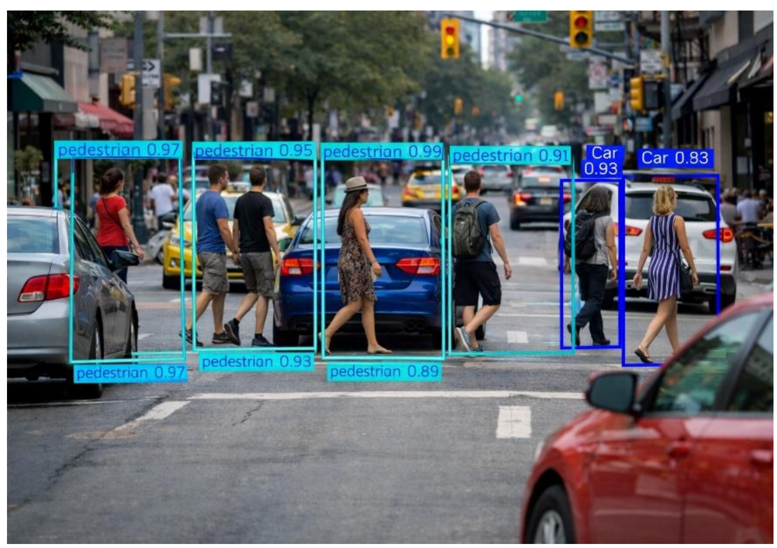
During testing, the trained YOLOv8 model performed well in most scenarios. However, some limitations were observed in challenging real-world conditions.

- A few false positives occurred where background objects were mistakenly detected as cars or pedestrians.
- Some false negatives were observed when small or partially hidden pedestrians were missed by the model.
- Detection performance decreased in images with complex backgrounds or low lighting conditions.
- The model occasionally struggled with distant or occluded objects, leading to inaccurate localization.
- In certain cases, class misclassification occurred due to visual similarities between objects.
- Detection performance may have been affected due to annotation inconsistencies, as some images contained mixed segmentation and bounding box labels, which were removed during training.

These observations highlight the challenges faced by the model in detecting small, occluded, or visually complex objects in real-world environments.

1.False Negative:

- pedestrian missed
- car missed

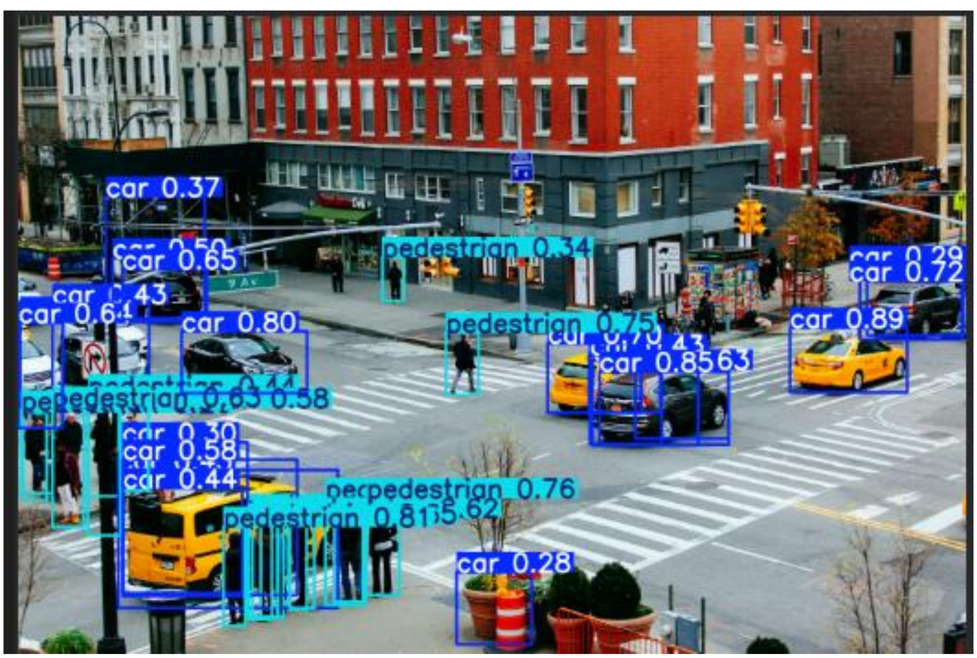


2. False Positive

- Bounding box drawn but no object there

3. Crowded Scene / Occlusion

- Too many objects
- Incorrect or overlapping boxes



Approach and Improvements Attempted :

- Applied data augmentation (brightness and blur) to improve robustness to lighting and image quality variations.
- Increased the effective training dataset size using 3× augmented images to enhance generalization.
- Data augmentation methods including brightness adjustment and Gaussian blur were implemented to simulate real-world environmental variations.
- Dataset preprocessing techniques such as auto-orientation and image resizing (640×640) were applied for consistent input formatting.
- Early stopping was applied during training to prevent overfitting and improve validation performance.
- Trained the model with higher epochs and used a patience parameter for optimal convergence.
- Evaluated the trained model on real-world unseen images to test practical performance.
- Adjusted the confidence threshold based on F1–Confidence curve to balance precision and recall.
- Dataset inconsistencies such as mixed segmentation and bounding box annotations were identified and corrected during the data preprocessing stage.

Conclusion:

In this project, a YOLO-based model was successfully trained to detect cars and pedestrians using a custom annotated dataset. The dataset was carefully prepared, preprocessed, and enhanced using augmentation techniques to improve learning and generalization.

The model achieved an overall mAP@0.5 of 0.769, which shows that it is able to correctly detect and localize most objects in the images. The precision and recall values were well balanced, indicating that the model performs reliably without producing excessive false detections or missing too many objects. Pedestrian detection performed slightly better than car detection.

Through detailed evaluation using training graphs, confusion matrix, and performance curves, it was observed that the model works well in normal conditions. However, some limitations were found in challenging situations such as small objects, occlusion, and complex backgrounds. These observations helped identify areas where the model can be further improved.

Overall, the model demonstrates stable training behavior, good generalization, and satisfactory real-world detection capability, making it a strong foundation for pedestrian and vehicle detection tasks.